

2/7


```
def fetch_director(obj):
    L=[]
    for i in ast.literal_eval(obj):
        if i['job']=='Director':
            L.append(i['name'])
            break
    return L
```

```
movies['crew']=movies['crew'].apply(fetch_director)
```

```
movies.head(1)
```

	movie_id	title	overview	genres	keywords	cast	crew
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[Action, Adventure, Fantasy, Science Fiction]	[culture clash, future, space war, space colon...	[Sam Worthington, Zoe Saldana, Sigourney Weaver]	[James Cameron]

so we have finally got what we wanted for our project by applying sum preprocessing step

But before that we have a problem that keywords ,cast,crew, and overview have space in words so we will have to remove the sapace and join them

now we will also convert the overview columns to a list so that we can concat all the list and we have a coloumns named tag

```
movies['overview']=movies['overview'].apply(lambda x:x.split())
```

we are applying a lambda function which give x and in that we have list comprehension that replace space with nothing

```
movies['genres']=movies['genres'].apply(lambda x:[i.replace(" ", "") for i in x])
movies['keywords']=movies['keywords'].apply(lambda x:[i.replace(" ", "") for i in x])
movies['cast']=movies['cast'].apply(lambda x:[i.replace(" ", "") for i in x])
movies['crew']=movies['crew'].apply(lambda x:[i.replace(" ", "") for i in x])
```

```
movies.head(5)
```

	movie_id	title	overview	genres	keywords	cast	crew
0	19995	Avatar	[In, the, 22nd, century, a, paraplegic, Marin...	[Action, Adventure, Fantasy, ScienceFiction]	[cultureclash, future, spacewar, spacecolony, ...	[SamWorthington, ZoeSaldana, SigourneyWeaver]	[JamesCameron]
1	285	Pirates of the Caribbean: At World's End	[Captain, Barbossa,, long, believed, to, be, d...	[Adventure, Fantasy, Action]	[ocean, drugabuse, exoticisland, eastindiatrad...	[JohnnyDepp, OrlandoBloom, KeiraKnightley]	[GoreVerbinski]
2	206647	Spectre	[A, cryptic, message, from Bond's past	[Action, Adventure,	[spy, basedonnovel, secretagent sequel mif	[DanielCraig, ChristophWaltz	[SamMendes]

```
import re
```

```
def remove_spaces_from_list_elements(data):
    """
    Removes all spaces from each element within a list.
    If the input is not a list, it returns the original data.
    """
    if isinstance(data, list):
        return [re.sub(r'\s+', '', item) for item in data]
    return data
```

```
# Apply to your DataFrame columns
movies['genres'] = movies['genres'].apply(remove_spaces_from_list_elements)
movies['keywords'] = movies['keywords'].apply(remove_spaces_from_list_elements)
movies['cast'] = movies['cast'].apply(remove_spaces_from_list_elements)
movies['crew'] = movies['crew'].apply(remove_spaces_from_list_elements)
```

```
movies['tags']=movies['overview']+movies['genres']+movies['keywords']+movies['cast']+movies['crew']
```

```
new_df=movies[['movie_id','title','tags']]
```

```
new_df.head(5)
```

	movie_id		title	tags
0	19995		Avatar	[In, the, 22nd, century,, a, paraplegic, Marin...
1	285	Pirates of the Caribbean: At World's End		[Captain, Barbossa,, long, believed, to, be, d...
2	206647		Spectre	[A, cryptic, message, from, Bond's, past, send...
3	49026	The Dark Knight Rises		[Following, the, death, of, District, Attorney...
4	49529		John Carter	[John, Carter, is, a, war-weary,, former, mili...

now after concating the columns we will convert the list back to string

```
new_df['tags']=new_df['tags'].apply(lambda x:" ".join(x))
```

```
<ipython-input-75-53a061086f2e>:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
new_df['tags']=new_df['tags'].apply(lambda x:" ".join(x))
```

```
new_df.head(5)
```

	movie_id		title	tags
0	19995		Avatar	In the 22nd century, a paraplegic Marine is di...
1	285	Pirates of the Caribbean: At World's End		Captain Barbossa, long believed to be dead, ha...
2	206647		Spectre	A cryptic message from Bond's past sends him o...
3	49026	The Dark Knight Rises		Following the death of District Attorney Harve...
4	49529		John Carter	John Carter is a war-weary, former military ca...

```
# converting tags to lower
```

```
new_df['tags']=new_df['tags'].apply(lambda x:x.lower())
```

```
<ipython-input-77-10b562b6e8d2>:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
new_df['tags']=new_df['tags'].apply(lambda x:x.lower())
```

```
new_df.head(5)
```

	movie_id		title	tags
0	19995		Avatar	in the 22nd century, a paraplegic marine is di...
1	285	Pirates of the Caribbean: At World's End		captain barbossa, long believed to be dead, ha...
2	206647		Spectre	a cryptic message from bond's past sends him o...
3	49026	The Dark Knight Rises		following the death of district attorney harve...
4	49529		John Carter	john carter is a war-weary, former military ca...

Performing Steaming on the corpus

```
import nltk
from nltk.stem.porter import PorterStemmer
ps=PorterStemmer()
```

```
def stem(text):
    y=[]
    for i in text.split():
        y.append(ps.stem(i))
    return " ".join(y)
```

```
new_df['tags']=new_df['tags'].apply(stem)
```

```
>ipython-input-83-402dd7cb6b10>:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
new_df['tags']=new_df['tags'].apply(stem)
```

****Now we will perform Text Vectorization ****

```
from sklearn.feature_extraction.text import CountVectorizer
cv=CountVectorizer(max_features=5000,stop_words='english')
```

```
cv.fit_transform(new_df['tags']).toarray().shape
```

```
(4806, 5000)
```

```
vectors=cv.fit_transform(new_df['tags']).toarray()
```

```
vectors
```

```
array([[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       ...,
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0]])
```

```
feature_names = cv.get_feature_names_out()
print(feature_names)
```

```
['000' '007' '10' ... 'zone' 'zoo' 'zoeydeschanel']
```

```
from sklearn.metrics.pairwise import cosine_similarity
```

```
similarity = cosine_similarity(vectors)
```

```
similarity[1]
```

```
array([0.08346223, 1.          , 0.06063391, ..., 0.02378257, 0.
       0.02615329])
```

```
#let fetch the index of the movie
new_df[new_df['title']=='The Lego Movie'].index[0]
```

```
744
```

```
# aafter having the movie index we are using that index and fetching similiarity vectors
print(similarity[0])
print(similarity[new_df[new_df['title']=='The Lego Movie'].index[0]])
```

```
[1.          0.08346223 0.0860309 ... 0.04499213 0.
 0.06488857 0.06859943 0.03535534 ... 0.02773501 0.
]
```

```
# after having similarity now we are sorting,so as soon as we are sorting we are loosing the index position
# so we are using enumerate function
list(enumerate(similarity[0]))
sorted(list(enumerate(similarity[0])), reverse=True, key=lambda x: x[1])[1:6]
```

```
[(1214, 0.28676966733820225),
 (2405, 0.26901379342448517),
 (3728, 0.2605130246476754),
 (507, 0.255608593705383),
 (539, 0.25038669783359574)]
```

```
def recommend(movie):
    movie_index = new_df[new_df['title'] == movie].index[0]
    distances = similarity[movie_index]
    movies_list = sorted(list(enumerate(distances)), reverse=True, key=lambda x: x[1])[1:6]

    for i in movies_list:
        print(new_df.iloc[i[0]].title)

recommend('Batman Begins')
```

→ The Dark Knight
Batman
Batman
The Dark Knight Rises
10th & Wolf

So we have our Recomodation system redy now we need to convert to the website

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